



Bubble Sort: Khel Khel Mein Sorting!



1. Sorting Kya Hai?

Socho tumhare paas bohot saare khilone (toys) alag-alag size ke hain. Agar tum unhe sabse chhote se lekar sabse bade ke line mein laga do, toh usko **Sorting** kehte hain! 📏

Programming mein, numbers ya names ko *Increasing* (badhte kram) ya *Decreasing* (ghat-te kram) order mein lagane ko Sorting kehte hain.



2. Bubble Sort Kya Hota Hai?

Bubble Sort sabse aasan sorting technique hai. Isme hum kya karte hain?

- Hum line se **do padosi (consecutive)** elements ko check karte hain.
- Agar pehla wala bada hai aur dusra chhota, toh hum unki jagah badal dete hain (Swapping). 🔄
- Aise karte-karte, sabse bada number paani ke bulbule (bubble) ki tarah tair kar sabse end mein chala jata hai! 🫧



3. Python Mein Swapping (Magic Trick!)

C++ ya Java mein swapping ke liye ek naya variable (T) lena padta tha (Jaise 3 glass ka khel). Par Python mein yeh ek jadu line se ho jata hai:

```
a, b = b, a # Bas itna likho aur jagah badal gayi! ✨
```



4. Step-by-Step Example

Maan lo hamare paas yeh numbers hain: 64, 32, 25, 45, 20, 15

Step 1 (Pehli Iteration): Hum shuruwat ke 2 numbers dekhenge.



Kya $64 > 32$ hai? Haa! Toh dono ki jagah badal do (Swap).



Ab 64 aur 25 ko compare karenge. Phir swap hoga. Aise hi aage badhte jayenge. Jab poori line ek baar check ho jayegi, toh sabse bada number (64) apni sahi jagah (sabse end mein) pahunch jayega!



Ise hum **First Iteration** kehte hain. 64 game se bahar ho gaya (wo set hai!). Ab bache hue numbers par yahi process repeat karenge, jab tak sab sahi jagah na aa jayein.



5. Python Code for Bubble Sort

Bohot asaan sa code hai, dhyan se dekho:

```
def bubble_sort(A):
    n = len(A)
    # i iteration count karega
    for i in range(n - 1):
        # j padosi numbers ko check karega
        # (n - 1 - i) kyu? Kyunki last ke numbers already set ho
        chuke hain!
        for j in range(n - 1 - i):
            if A[j] > A[j+1]:
                # Swap kar do! 🔄
                A[j], A[j+1] = A[j+1], A[j]

# Test karte hain!
arr = [64, 32, 25, 45, 20, 15]
bubble_sort(arr)
print("Sorted Array:", arr)
```



Important Points to Remember!

- **Iterations:** Agar humare paas n elements hain, toh max $n-1$ iterations lagti hain.
- **Nested Loops:** Bubble sort mein humesha do (2) loops use hote hain. Ek iteration ke liye (outer loop), aur ek padosi check karne ke liye (inner loop).
- **Time Complexity:** Iski Time Complexity $O(n^2)$ hoti hai. Kyunki loop ke andar loop (nested loops) n times chal raha hai.
- **Swapping:** Python mein swap karna sabse aasan hai: $x, y = y, x$.

Happy Coding! Keep learning everyday! 🚀